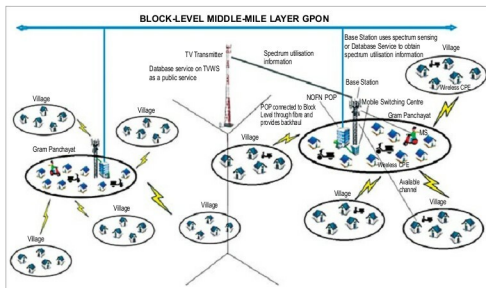


A model for rural broadband connectivity

The proposed model requires the MSC to sense or utilise geo-location database service to receive available spectrum information. A multi-user channel-allocation protocol is required to support fair spectrum allocation to neighboring *gram panchayats* and also among end users under a *gram panchayat*. The micro-cell concept and the co-existence strategy between primary and secondary users are the key enabling factors in this model.

Challenges in implementing CR-TVWS in India

The socio-economic consequences of allocating TVWS using various options such as unlicensed, auction, controlled, licensed shared access (LSA), authorised shared access (ASA), etc have to be carefully evaluated.



Proposed model to provide last-mile connectivity to villages using CRN based on TVWS

Standards for communications in CRN have to be laid down; transmitting power of customer premises equipment, mobile equipment and base stations in the secondary network will have to be laid down to avoid harmful interference to incumbents' network. IEEE 802.22 WRAN (wireless regional area network) standard is defined for accessing licensed TV-band channels by secondary users on a non-interference basis.

Rules need to be formulated to protect other sensitive networks such as military, radio astronomy and healthcare services operating in the same band. A detailed guideline is required on allocation, responsibilities and operation of geo-location database services. There is no

Policies to grant licences to use the TV band spectrum to run experimental networks need to be framed. Data regarding TV bands used in each region in the country is required so that interference levels can be checked. There is no easily-available data regarding location of TV transmitters in the country.

In addition to the challenges depicted above, successful deployment of CRN requires well-planned and coordinated functioning of multiple layers of communications. However, technical aspects of communication are beyond the scope of this article and hence not discussed. **EFY**